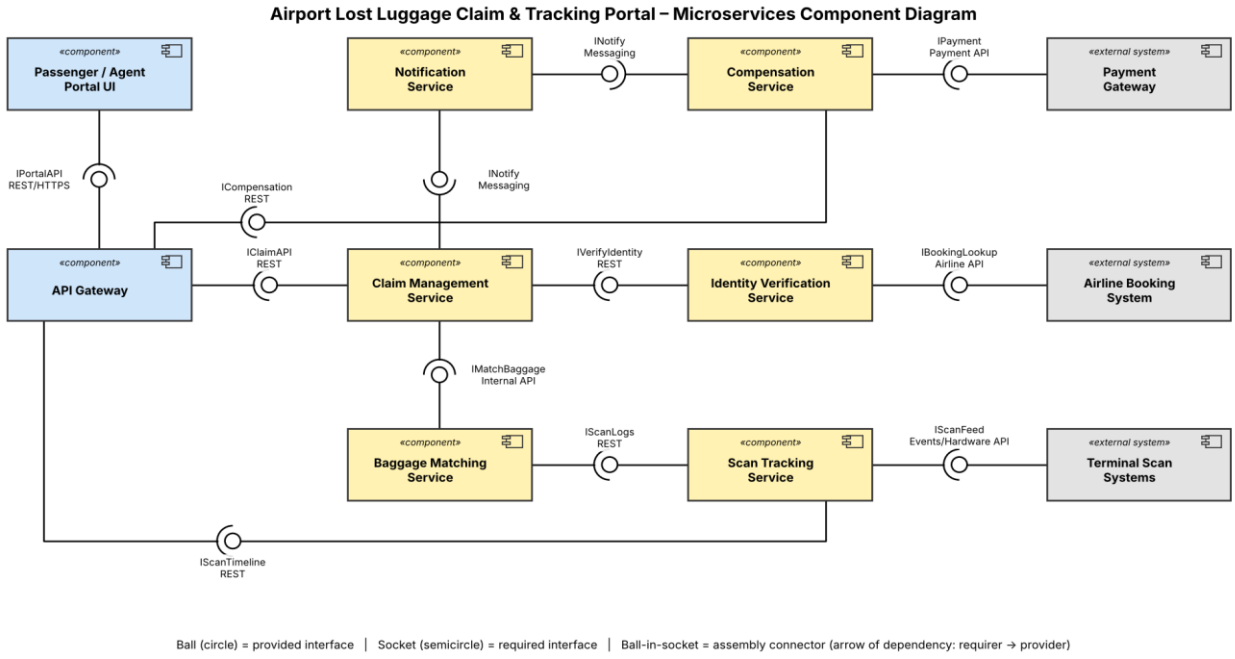
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SE LAB 3



Architecture Selection: I chose Microservices Architecture for the Airport Lost Luggage Claim & Tracking Portal.

Architectural Choice: Microservices. The system is split into independently deployable services: Claim Management, Identity Verification, Baggage Matching, Scan Tracking, Compensation and Notification. An API Gateway and a Portal UI sit in front of them.

Reason 1: Availability during peak travel (NFR-002). Baggage loss peaks exactly when travel is busiest, and claim submission must stay at 99.9% uptime. With microservices, the Claim Management Service can be replicated and scaled on its own. A failure in Compensation or Notification does not stop passengers from filing claims, whereas in a Layered system one overloaded layer could take down the whole portal.

Reason 2: Different workloads and independent evolution. Scan Tracking handles a continuous stream of events from many terminals. Compensation is a rare, SLA-timer-driven task, and Matching (FR-001) is compute-heavy. Separate services let each be scaled, deployed and changed independently. For example, the payment provider can be swapped without redeploying the claim flow. They also communicate only through the defined interfaces (IClaimAPI, IMatchBaggage, IScanLogs, etc.).

Security Advantage: All external traffic enters through the API Gateway, which authenticates requests and enforces access control. Passenger and agent roles are kept separate. Identity Verification is isolated in its own service, so personal data is confined to a small, auditable component. Payment details are handled only by Compensation and the external Payment Gateway. A compromise of one service does not expose the others.

Performance Benefit: The Scan Tracking Service holds the scan history and serves the timeline on its own, so it can be scaled horizontally and cached independently. This helps meet the under-200 ms timeline target (NFR-001). Matching runs separately from request handling, so heavy cross-referencing does not slow passenger-facing calls.